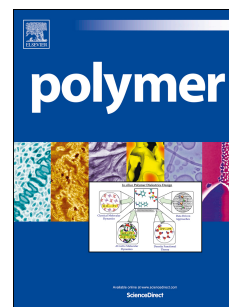


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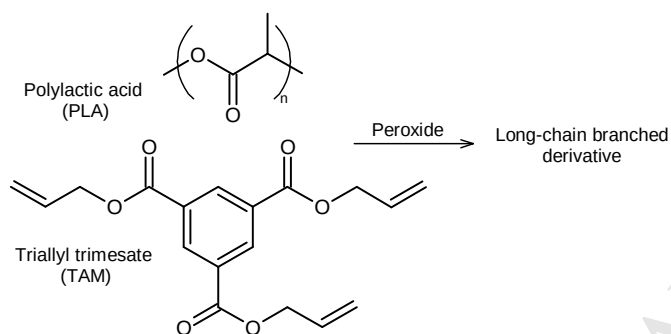
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**Peroxide-initiated Graft Modification of Thermoplastic BioPolyesters:****Introduction of Long-Chain Branching**

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**Graphical Abstract**

**Peroxide-initiated Graft Modification of Thermoplastic BioPolyesters:****Introduction of Long-Chain Branching**

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**Abstract**

The chemical modification of polylactic acid (PLA) and related polyesters by peroxide-initiated grafting of multifunctional allylic and acrylic monomers is described. Solvent-free, melt-state reactions of PLA and triallyl trimesate (TAM) are shown to be remarkably efficient compared to analogous reactions of polyolefins such as poly(ethylene-co-octene) (EOC), requiring very low reagent loadings to produce a long chain branched (LCB) architecture. A systematic study of potential transesterification chemistry confirms that ionic reactions do not contribute significantly to PLA reactivity under typical grafting conditions. Further explorations of H-atom abstraction and monomer addition yields demonstrate that PLA is relatively unreactive toward the radical reactions that support LCB production. Careful examination of the solubility of coagent intermediates reveal dramatic differences in oligomer solubility between a hydrocarbon system and an ester-based matrix. Whereas TAM conversion in hydrocarbon leads to a precipitation polymerization that removes coagent from solution, reactions conducted in the polyester do not incur losses of oligomeric intermediates, indicating that the ability of PLA to retain multi-functional intermediates is the underlying cause of its remarkable LCB grafting yields.

**Keywords**

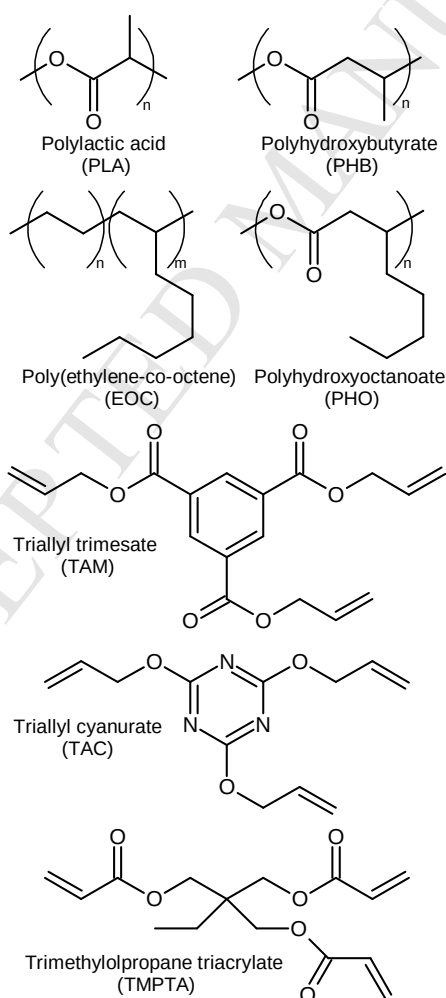
Polymer modification, long chain branching, polylactic acid

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## 1. Introduction

Poly(lactic acid) (PLA, Scheme 1) is a biodegradable thermoplastic polyester that can be synthesized from starch-based feedstocks, making it an attractive alternative to petroleum based plastics [1,2]. However, the material has notable deficiencies such as low melt strength, pronounced brittleness [3,4], and slow rates of crystallization from the melt state [5,6]. This has hindered application development efforts on this material, limiting it to products where bio-sourcing and biodegradability are paramount [7].

Several strategies for improving PLA properties have been explored, including compounding with low molecular weight additives and/or polymers, copolymerizing lactic acid with alternative monomers, [6–12] and chemical modification of the homopolymer using ultra violet, gamma ray, or electron beam irradiation [13–19]. Of principal interest to this work is the graft modification of PLA with organic peroxide formulations in a solvent-free, reactive melt compounding approach [20–23].



Scheme 1: Polymers and coagents used in this work

Takamura et al. [24,25] examined the effect various peroxides on PLA, demonstrating the tendency of this polymer to crosslink under the action of a radical initiator. The long chain branching (LCB) generated by this chemistry was shown to accelerate crystallization by affecting PLA crystallite nucleation. Wang et al. [16] and Fang et al. [17] examined the graft modification of PLA with a trifunctional monomer, trimethylolpropane triacrylate (TMPTA) using radiation-induced macroradical generation. In both cases, evidence of LCB was confirmed by melt-state rheology, as the chemically-modified derivatives showed more intense shear thinning character and heightened elasticity at low shear rates. Similar results have been reported by You et al. [20], who used dicumyl peroxide (DCP) to initiate the addition of pentaerythritol triacrylate to PLA. As a result of a comb-like branch architecture, their reaction products demonstrated higher crystallization temperatures, and improved nucleation density during crystallization.

More recent research has revealed the exceptional sensitivity of PLA to coagent-based formulations [22,26], wherein relatively small amounts of allylic reagents such as triallyl trimesate (TAM, Scheme 1) generate much higher LCB contents than observed for similar polyolefin modifications [21,23]. The objective of this work was to determine the underlying cause of this remarkable efficacy. After a brief comparison of PLA modifications to polyolefin reactions, the study shifts to an assessment of potential ionic and radical reaction pathways. Measurements of monomer graft yields, H-atom transfer yields, melt-state rheology, and molecular weight distribution are discussed in the context of LCB generation in the PLA system and related polyesters, including polyhydroxybutyrate (PHB) and polyhydroxyoctanoate (PHO). The work concludes with model compound studies of the role of coagent oligomer solubility in LCB reactions.

## 2. Materials and Methods

### 2.1 Materials

An extrusion / thermoforming grade of polylactic acid (PLA, NatureWorks® grade 2500 HP, MFI =  $8.0\text{g}\cdot 10\text{min}^{-1}$  at  $190^\circ\text{C}$ ) was purified by dissolution / precipitation (chloroform/methanol) and drying under vacuum at  $60^\circ\text{C}$ . Poly(ethylene-co-octane) (EOC, Dow Chemical grade 8200, 10 mol% octane [27], MFI  $5.0\text{g}\cdot 10\text{min}^{-1}$  at  $190^\circ\text{C}$ ) was used as received, as was additive-free polyhydroxybutyrate (PHB, Biomer) with a weight-average molecular weight of  $1416\text{ kg}\cdot\text{mol}^{-1}$  and a dispersity of 7.0. Polyhydroxyoctanoate (PHO) containing 98 mol% 3-hydroxyoctanoate and 2 mol% 3-hydroxyhexanoate, was produced in house through a chemostat culture with addition of acrylic acid to inhibit  $\beta$ -oxidation [28]. Butyl acrylate (BA, 99%), dicumyl peroxide (DCP, 98%), triallyl cyanurate (TAC, 97%) and trimethylolpropane triacrylate (TMPTA, 70%) were used as received from Sigma Aldrich, while allyl benzoate (AB, 98%) and triallyl trimesate (TAM, 98%) were used as supplied by TCI America and Monomer-Polymer Labs, respectively.

## 2.2 Compounding

Samples for graft modification with monofunctional monomers AB and BA were prepared by coating the desired polymer (0.5 g) with an acetone solution to create a masterbatch containing DCP (0.001 g,  $18.5 \mu\text{mol}\cdot\text{g}^{-1}$ ), and monomer (0.05g,  $308.6 \mu\text{mol}\cdot\text{g}^{-1}$ ). After evaporation of acetone, masterbatches were charged to an Atlas Mixer at  $180^\circ\text{C}$  for 6 min. The products were purified by dissolution/precipitation and dried prior to further analysis.

Compounds for trifunctional coagent graft modification were prepared by coating polymer (10 g) with a mixture of DCP (0.01 g,  $3.7 \mu\text{mol}\cdot\text{g}^{-1}$ ) and coagent (0.04 g,  $12.1 \mu\text{mol}\cdot\text{g}^{-1}$ ) in an acetone solution, and drying under vacuum at  $60^\circ\text{C}$  before processing with a twin-screw DSM micro-compounder at  $180^\circ\text{C}$  for 6 min at a screw speed of 100 RPM. Formulations are classified by starting material, the amounts of peroxide/coagent, and the coagent. For example, PLA 3.7/12.1 TAM denotes PLA reacted with  $[\text{DCP}] = 3.7 \mu\text{mol}\cdot\text{g}^{-1}$  and  $[\text{TAM}] = 12.1 \mu\text{mol}\cdot\text{g}^{-1}$ .

## 2.3 Rheology

Compression-molded discs (25 mm diameter, 2 mm thick) were prepared on a Carver press at  $180^\circ\text{C}$  and dried under vacuum prior to analysis. Linear viscoelastic properties were measured in oscillatory shear mode at  $180^\circ\text{C}$  using an MCR-301 Anton Paar rheometer equipped with a 25 mm parallel plate. The frequency range was between  $0.05 - 100 \text{ rad}\cdot\text{s}^{-1}$ , with lower frequencies avoided to limit opportunity for polymer degradation.

## 2.4 Gel Permeation Chromatography (GPC)

GPC analysis of THF solutions was conducted using a Viscotek 270 max instrument equipped with differential refractive index (DRI), viscosity (IV), and light scattering detectors. The separation module was maintained at  $40^\circ\text{C}$  and contained two porous PolyAnalytik columns in series with an exclusion molecular weight limit of  $20 \times 10^6$  Da. The data was processed with Viscotek Omnisec software using  $dn/dc$  values of 0.0482 for PLA in THF [29] and 0.0788 for EOC in THF, the latter determined by analysis of EOC+THF solutions ( $0.2\text{--}1.0 \text{ mg}\cdot\text{mL}^{-1}$ ) with an Wyatt Optilab DSP refractometer operating at 690 nm.

## 2.5 Gel Content

Gel contents were measured according to ASTM D2765 by extracting polymer samples from stainless-steel wire mesh (120 sieve) for 8 h with boiling THF and drying under vacuum at  $60^\circ\text{C}$  to constant weight. Data are reported as the weight percent of unextracted material.

## 2.6 Nuclear Magnetic Resonance (NMR) Spectroscopy

$^1\text{H}$  NMR spectra were acquired with a Bruker AC-400 MHz spectrometer in  $\text{CDCl}_3$  using polymer concentrations of  $10\text{ mg}\cdot\text{mL}^{-1}$ , with chemical shifts referenced to residual  $\text{CHCl}_3$ .

### 2.7 Abstraction Efficiency

Finely-ground polymer (0.5 g) was coated with a stock solution of DCP in acetone ( $\sim 0.2\text{ mL}$ ) to produce a peroxide concentration of 2 wt% ( $74.0\text{ }\mu\text{mol}\cdot\text{g}^{-1}$ ). The acetone was allowed to evaporate before charging the mixture to the Atlas Laboratory Mixer at  $180^\circ\text{C}$  for 6 min. The resulting material was dissolved in acetone and a small aliquot of the solution was injected into a Hewlett-Packard 5890 series II chromatograph equipped with a Superlco SPB-1 microbore column using  $2\text{ mL}\cdot\text{min}^{-1}$  of helium as carrier gas. Injector and detector temperatures were held at  $275^\circ\text{C}$ , with the oven temperature starting at  $100^\circ\text{C}$  for 2 min, ramping to  $250^\circ\text{C}$  at  $22^\circ\text{C}\cdot\text{min}^{-1}$ , and holding for 6 min. Abstraction efficiencies are reported as  $[\text{cumyl alcohol}] / ([\text{cumyl alcohol}] + [\text{acetophenone}])$ .

## 3. Results

### 3.1 Long Chain Branching Efficiency of Coagent-based PLA Modifications

Earlier reports of peroxide-initiated PLA modification have described the susceptibility of this thermoplastic toward crosslinking when it is subjected to radical chemistry, presumably through H-atom abstraction from the polymer by initiator-derived radicals, followed by combination of the resulting macroradical intermediates. This conventional peroxide crosslinking sequence is well-documented for ethylene-octene copolymers (EOC) [30–35] and, as such, EOC is a valuable benchmark for assessing PLA branching efficiency. The grade of EOC chosen as a reference had melt-state rheological properties similar to those of our PLA starting material. Figure 1 a-c provide complex viscosity, storage modulus, and phase angle data for EOC and its derivatives, while Figure 1 d-f illustrate the same measurements for the PLA system.

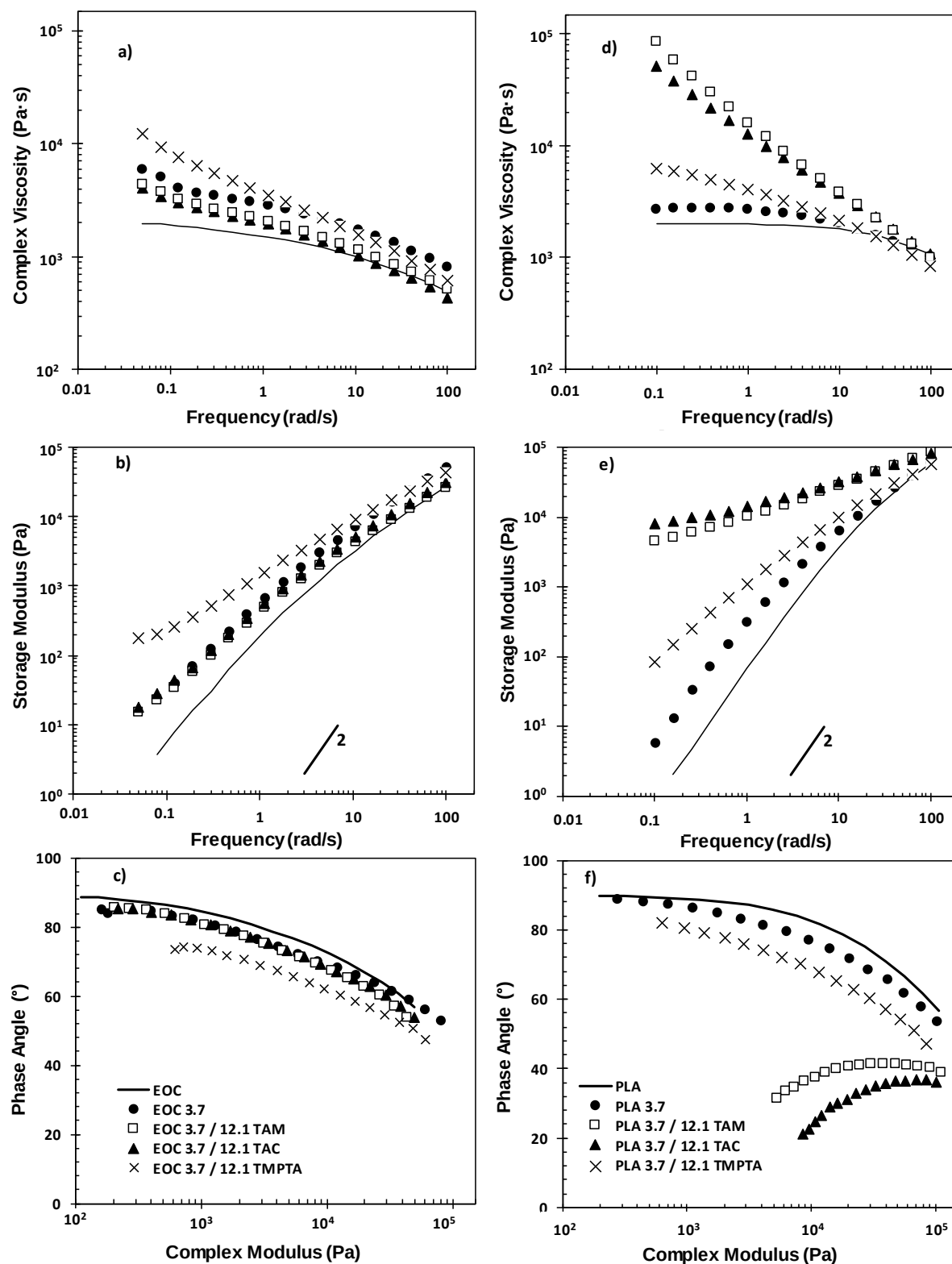


Figure 1: (a,d) complex viscosity and (b,e) storage modulus as a function of frequency, (c,f) van Gurp-Palmen plot for EOC (a-c) and PLA (d-f) formulations, respectively ( $T=180^{\circ}\text{C}$ ).

Heating both polymers with organic peroxide affected zero-shear viscosity and degree of shear thinning, but to different extents. Treatment of EOC with a peroxide loading of  $3.7 \mu\text{mol}\cdot\text{g}^{-1}$  eliminated the Newtonian plateau observed for the starting material (Figure 1a), whereas the same treatment of PLA produced only a slight increase in the zero-shear viscosity (Figure 1d). Insensitivity to peroxide-initiated modification can be attributed to low yields of H-atom abstraction from the polymer and/or differences in the extent of macroradical termination by combination as opposed to disproportionation. The reactivity of PLA in H-atom transfer reactions is examined in Section 3.3.

Although PLA was relatively unresponsive to peroxide-only formulations, its sensitivity to multifunctional monomers far exceeded that of EOC, particularly for the allylic coagents triallyl trimesate (TAM) and triallyl cyanurate (TAC). Consider the complex viscosity data plotted in Figures 1 a,d that reveal dramatic increases in PLA melt viscosity for DCP+TAM and DCP+TAC formulations, with all evidence of a Newtonian plateau lost in favour of a power-law relationship, and deviations in terminal flow behavior (Figures 1 b,e). These changes, along with the shifts in low frequency storage modulus data plotted in Figures 1b and 1e, provide unambiguous evidence of the heightened reactivity of PLA to radical-mediated coagent modification.

Insight into the architecture of coagent-modified PLA is provided by the van Gorp-Palmen plots of phase angle against complex modulus (Figures 1c,f). The EOC and PLA starting materials demonstrated phase angles that approach  $90^\circ$  in the low modulus region, as is expected for linear polymers with limited chain entanglement. Long chain branching created by coagent grafting changes the relaxation of chains when subjected to an oscillatory deformation, enhancing elasticity at low frequency / complex modulus [36,37]. This is evidenced by phase angle reductions, which are particularly pronounced for PLA derivatives prepared from the allylic coagents.

Gel content and molecular weight distribution measurements provided additional information regarding changes to polymer chain architecture. The data presented in Table 1 show that none of the formulations produced a measurable gel fraction from EOC. However, all the EOC derivatives had higher molecular weight averages than the starting material, as expected from the melt-state rheology data detailed above. In the case of PLA, peroxide alone had little effect on number-average or weight-average molecular weights. However, the combination of allylic coagent and peroxide produced substantial gel fractions that precluded GPC analysis, but is consistent with the heightened melt elasticity seen for these materials.

Table 1: Gel content and molecular weight data for unmodified polymers and their derivatives.

| Polymer | DCP Loading ( $\mu\text{mol}\cdot\text{g}^{-1}$ ) | Coagent Loading ( $\mu\text{mol}\cdot\text{g}^{-1}$ ) | Coagent | Gel Content (wt %) | $M_n$ ( $\text{kg}\cdot\text{mol}^{-1}$ ) | $M_w$ ( $\text{kg}\cdot\text{mol}^{-1}$ ) |
|---------|---------------------------------------------------|-------------------------------------------------------|---------|--------------------|-------------------------------------------|-------------------------------------------|
| EOC     | --                                                | Unmodified                                            | --      | 0                  | 36                                        | 165                                       |
| EOC     | 3.7                                               | 0.0                                                   | none    | 0                  | 44                                        | 181                                       |
| EOC     | 3.7                                               | 12.1                                                  | TAM     | 0                  | 49                                        | 178                                       |
| EOC     | 3.7                                               | 12.1                                                  | TAC     | 0                  | 47                                        | 174                                       |
| EOC     | 3.7                                               | 12.1                                                  | TMPTA   | 0                  | 54                                        | 230                                       |
| PLA     | --                                                | Unmodified                                            | --      | 0                  | 44                                        | 86                                        |
| PLA     | 3.7                                               | 0.0                                                   | none    | 0                  | 42                                        | 86                                        |
| PLA     | 3.7                                               | 12.1                                                  | TAM     | 18                 | gelled                                    | gelled                                    |
| PLA     | 3.7                                               | 12.1                                                  | TAC     | 21                 | gelled                                    | gelled                                    |
| PLA     | 3.7                                               | 12.1                                                  | TMPTA   | 0                  | 34                                        | 165                                       |

Lowering the coagent loading used in PLA modifications to  $4.5 \mu\text{mol}\cdot\text{g}^{-1}$  produced gel-free derivatives that were amenable to GPC analysis. Figure 2a provides representative molecular weight distributions and Figure 2b illustrates the corresponding light scattering detector responses. Whereas the PLA starting material and the product of the DCP-only formulation produced unimodal distributions, coagent-modified products showed complex light scattering data. Bimodal molecular weight and branching distributions are a well-established consequence of radical-mediated coagent modifications, having been documented extensively for polyolefin systems [5,26,38]. Briefly, those polymer chains that engage in coagent grafting become macromonomer derivatives with greater branching potential than unfunctionalized chains. As a result, reaction products are usually comprised of unmodified material and a population of high molecular weight, hyperbranched chains, the latter being responsible for observed changes in melt-state rheology and crystallization behaviour. Our GPC data show that the architecture of coagent-modified PLA is analogous to that that generated from polyolefins, with branching found predominately in the high molecular weight fraction of the chain population.

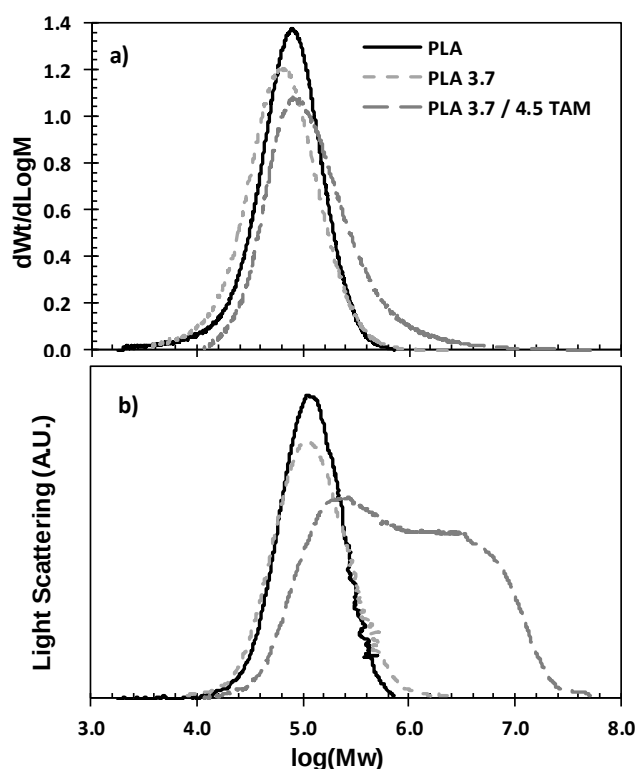


Figure 2: Molecular weight distributions and light scattering detector responses for PLA and its derivatives.

Significant interest in biodegradable polyesters such as polyhydroxybutyrate (PHB) and polyhydroxyoctanoate (PHO), motivated us to expand our study. The rheology data plotted in Figure 3 reveal interesting parallels to the EOC and PLA results described to this point. Of particular interest is the response to allylic coagents versus the acrylate-based coagent. Whereas PHB was highly sensitive to TAM and TAC, the response of PHO was considerably more subdued, with TMPTA proving to be most effective for this material. This suggests that the reactivity of PHB is closer to that of PLA, while the behaviour of PHO is more consistent with that of a polyolefin. The underlying cause of these relationships is addressed following a brief examination of the importance of non-radical pathways to polyolefin branching.

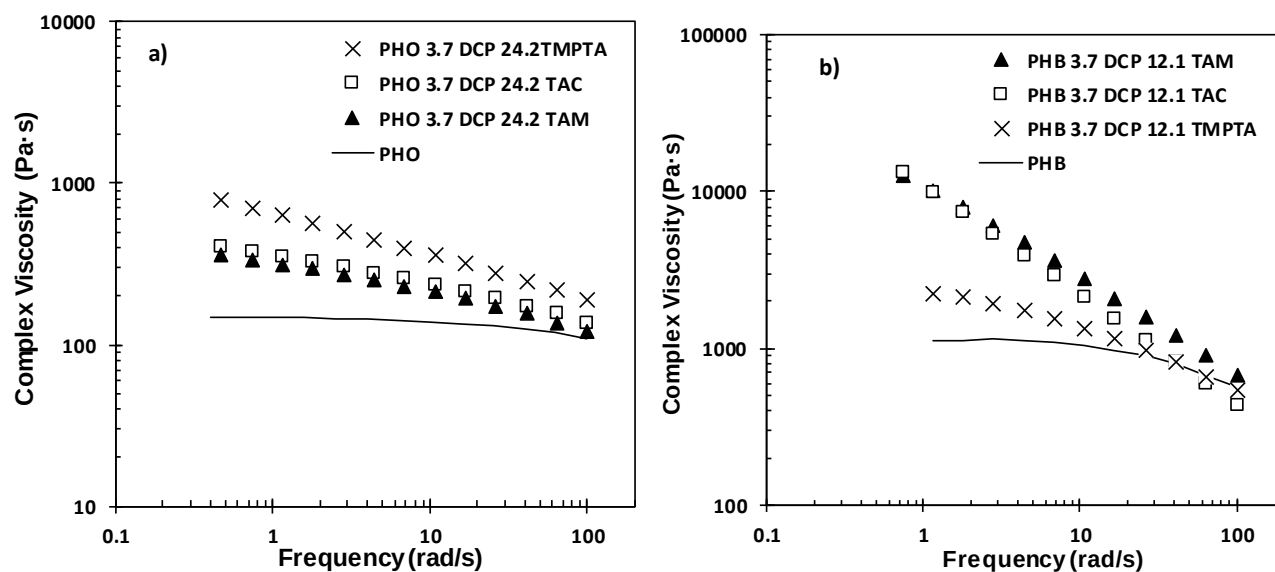


Figure 3: Complex viscosity as a function of oscillation frequency (a. PHO; b. PHB;  $T=180^{\circ}\text{C}$ ).

### 3.2 Contribution of Ionic Reactions to PLA Modification

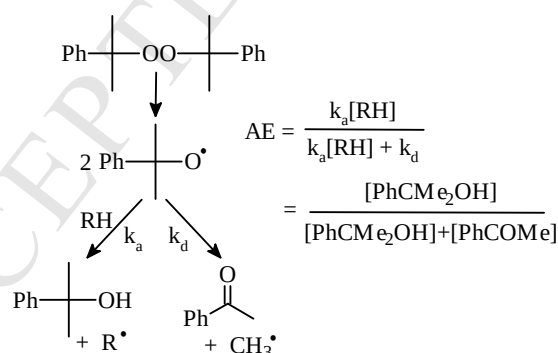
The susceptibility of PLA to hydrolysis is well documented [39–41], as is the potential for the material to engage in transesterification [11,42–45]. The latter may be relevant to this work, given the unusual efficiency observed for TAM in the PLA system. Note that our polymer modifications are conducted above the melting temperature of the thermoplastic and for relatively short time periods. If transesterification between PLA and coagent occurs under these conditions, it would produce a macromonomer derivative with heightened sensitivity to peroxide-initiated crosslinking. However, if transesterification is insignificant, then peroxide + TAM formulations operating on PLA would be limited to conventional radical grafting chemistry.

Rheology data acquired for the TAC system is noteworthy, as it indicates that the allyl cyanurate monomer, which is incapable of transesterification, is as effective as TAM in terms of PLA branching (Figure 1). More direct information regarding the intrinsic reactivity of PLA and TAM in the absence of peroxide was assessed by heating a sample containing  $12.1\ \mu\text{mol}\cdot\text{g}^{-1}$  of the coagent to  $180^{\circ}\text{C}$  for 6 min, and measuring the molecular weight distribution and the polymer-bound allylic ester content of the product. GPC analysis showed the molecular weight of the TAM-treated material ( $M_n=48\text{K}$ ,  $M_w=86\text{K}$ ) to be nearly unchanged from the PLA starting material ( $M_n=46\text{K}$ ,  $M_w=86\text{K}$ ). More importantly, NMR analysis of the purified product showed no evidence of polymer-bound allylic or aromatic functionality. Therefore, it can be concluded that the remarkable performance of allylic coagents with respect to PLA is not due to ionic reactions involving allyl group transfer.

### 3.3 Abstraction Efficiency and Monofunctional Coagent Graft Modification

Although the H-atom abstraction and monomer addition reactions that underlie the chemical modification of polyolefins have received considerable attention, very little is known about analogous PLA chemistry. A comprehensive understanding of coagent-based long chain branching (LCB) processes requires knowledge of the rates and regioselectivity of macroradical creation by H-atom transfer to initiator-derived radicals, both alkoxy and methyl, as well as the graft modification sequence involving C=C addition and H-atom abstraction by carbon-centred radicals derived from the monomer. To date, this level of detail has not been acquired for any polymer modification process. However, simple measurements of peroxide abstraction efficiency and monomer graft yields have provided considerable insight. Through comparison of PLA relative to a conventional polyolefin system such as EOC, the unique response of the polyester to allylic coagents can be better understood.

Abstraction efficiency (AE) is a measure of the fraction of cumyloxy radicals that abstract an H-atom from the polymer as opposed to undergoing fragmentation to a methyl radical (Scheme 2) [46]. It is determined by the yield of cumyl alcohol, the byproduct of H-atom transfer, and the yield of acetophenone, the byproduct of cumyloxy radical fragmentation, with  $AE = [\text{cumyl alcohol}] / ([\text{cumyl alcohol}] + [\text{acetophenone}])$ . Assuming that the rate of cumyloxy radical fragmentation is relatively insensitive to the reaction medium [47,48], AE becomes a quantitative measure of H-atom donation reactivity [49].



Scheme 2: DCP thermolysis byproducts and abstraction efficiency (AE)

Table 2 provides AE data recorded at 180°C for the four materials of interest, along with the concentration of H-atom-bearing functionality in each polymer. The PLA result of AE = 0.12 is somewhat surprising, given that the lowest value previously reported for this peroxide was for polyisobutylene at 0.13 [46]. Note that H-atom

transfer rates depend upon the amount and reactivity of each H-atom donor. The EOC used in this work was comprised of 90 mol% ethylene and 10 mol% octane, amounting to functional group contents of  $[-CH_2-] = 65 \text{ mmol}\cdot\text{g}^{-1}$ ,  $[-CH_3] = 2.7 \text{ mmol}\cdot\text{g}^{-1}$ , and  $[-CH-] = 2.7 \text{ mmol}\cdot\text{g}^{-1}$ . This abundance of potential H-atom donors in the polyolefin is in sharp contrast to PLA, which provided only  $[-CH_3] = 13.9 \text{ mmol}\cdot\text{g}^{-1}$ , and  $[-CH-] = 13.9 \text{ mmol}\cdot\text{g}^{-1}$ . Therefore, if AE is solely a function of the number of available sites for H-atom abstraction, PLA is expected to be the less reactive polymer.

Table 2: C-H bond content and radical reactivity of polymer substrates

| Polymer | $[-CH-]$<br>( $\text{mmol}\cdot\text{g}^{-1}$ ) | $[-CH_2-]$<br>( $\text{mmol}\cdot\text{g}^{-1}$ ) | $[-CH_3]$<br>( $\text{mmol}\cdot\text{g}^{-1}$ ) | Abstraction<br>Efficiency <sup>a</sup> | Allyl<br>Benzoate<br>Graft Yield<br>(%) <sup>b</sup> | Butyl<br>Acrylate<br>Graft Yield<br>(%) <sup>b</sup> |
|---------|-------------------------------------------------|---------------------------------------------------|--------------------------------------------------|----------------------------------------|------------------------------------------------------|------------------------------------------------------|
| EOC     | 2.7                                             | 65.0                                              | 2.7                                              | 0.68                                   | 35                                                   | 65                                                   |
| PLA     | 13.9                                            | 0.0                                               | 13.9                                             | 0.12                                   | 5                                                    | 5                                                    |
| PHO     | 7.0                                             | 32.2                                              | 7.0                                              | 0.60                                   | 40                                                   | 75                                                   |
| PHB     | 11.6                                            | 11.6                                              | 11.6                                             | 0.18                                   | -                                                    | -                                                    |

a.  $[\text{DCP}] = 74.0 \text{ }\mu\text{mol}\cdot\text{g}^{-1}$ ;  $180^\circ\text{C}$ .

b.  $[\text{DCP}] = 18.5 \text{ }\mu\text{mol}\cdot\text{g}^{-1}$ ;  $[\text{Monomer}] = 308 \text{ }\mu\text{mol}\cdot\text{g}^{-1}$ ;  $180^\circ\text{C}$ .

Fundamental studies of H-atom transfer rates have established the importance of both enthalpic and entropic effects for different H-atom donors. Where steric inhibition is not operative, such as in H-atom abstraction from methyl groups, homolytic bond dissociation energy (BDE) can dominate patterns of reactivity. The relatively high BDE of methyl groups reduces H-atom abstraction from this site, as evidenced by the low AE of polyisobutylene, and can be expected to contribute little to PLA reactivity. In contrast, the tertiary C-H position presents a relatively low BDE, owing to hyperconjugation with the adjacent methyl substituent, the potential lone-pair resonance effect of adjacent oxygen, and the potential inductive effect of the adjacent carbonyl – all acting to provide thermodynamic stability to a tertiary macroradical [50]. However, steric inhibition may act in opposition to these BDE effects, as recent experimental work, supported by quantum chemical calculations, have identified some H-atom transfers as entropy-controlled. Indeed, steric effects can outweigh enthalpic effects in tertiary H-atom donation, resulting in lower reactivity that is expected based on BDE arguments alone [51]. In the case of PLA, steric inhibition of the tertiary position, coupled with the relatively low abundance of tertiary groups, can account for its observed AE.

The H-atom transfer reactivity of the other polyesters is consistent with their composition, with the AE = 0.60 for PHO correlating with its high aliphatic hydrocarbon content, and the more moderate AE=0.18 for PHB being consistent with its limited C-H bond donor concentration.

The low H-atom transfer reactivity of PLA accounts for its marginal response to a peroxide-only LCB process, in that the extent of crosslinking is proportional to macroradical yields generated by initiator thermolysis products. However, its exceptional response to allylic coagents is more difficult to explain, given the close relationship previously reported between AE values and the yield of monomer grafting processes. Studies of vinyltrialkoxysilane addition to various polymers have proven that graft yields correlated strongly with AE, and this relationship was justified on the basis that the monomer grafting sequence involves a H-atom transfer process [34,52,53].

To evaluate the reactivity of PLA toward monomer addition, the yields of peroxide-initiated allyl benzoate (AB) and butyl acrylate (BA) were measured as monofunctional analogues to TAM and TMPTA, respectively. The data listed in Table 2 show that PLA is relatively unreactive with respect to monomer grafting, producing reaction yields that are a fraction of those generated by EOC and PHO. Unfortunately, the insolubility of PHB in common NMR solvents precluded its analysis. Nevertheless, this monomer conversion data proves that the sensitivity of PLA toward coagent branching cannot be attributed to an efficient grafting process.

### 3.4 Solubility Considerations

The experiments described above have shown that the efficacy of allylic coagents toward PLA is not due to ionic reactions such as transesterification, nor is it attributable to an efficient radical grafting sequence. A third potential factor is the solubility of branched intermediates within the polymer matrix. Careful studies of polypropylene modification with TMPTA and TAM have shown that a precipitation polymerization mechanism operates concurrently with polymer branching during a peroxide-initiated process [54]. Furthermore, small molecule studies have confirmed that an initially homogeneous condition produces highly crosslinked particles that when alkane solutions of TAM are heated with DCP [55]. This crosslinked phase results from the precipitation of coagent oligomers from solution to generate a monomer-rich phase whose continued crosslinking produces insoluble particles. The result is a diversion of coagent from polyolefin crosslinking toward the generation of a precipitated phase.

An examination of Hansen solubility parameters used to assess the miscibility of polymers and solvents is revealing. The reported dispersion, polar and hydrogen-bonding parameters for polyethylene ( $\delta_D = 16.9$ ,  $\delta_P = 0.8$ ,  $\delta_H = 2.8$ ) are consistent with its saturated hydrocarbon composition, while those of PLA ( $\delta_D = 18.6$ ,  $\delta_P = 9.9$ ,  $\delta_H = 6.0$ ) are indicative of the polarity of the polyester and its capacity for hydrogen bonding. Although Hansen parameters for TAM and TMPTA are unavailable, values for the difunctional analogues, diallyl phthalate

( $\delta_D = 22.2$ ,  $\delta_P = 12.2$ ,  $\delta_H = 8.6$ ) and 1,4-butanediyl diacrylate ( $\delta_D = 16.8$ ,  $\delta_P = 9.1$ ,  $\delta_H = 4.2$ ), provide insight into the phase equilibrium behaviour of allylic aromatic esters and acrylate monomers.

These values indicate that PLA has a much greater thermodynamic affinity for coagents than does EOC, thereby supporting the solubility of coagent oligomerization products. This is an important element of the branching process, since initiation of coagent oligomerization by alkyl radical addition, followed by precipitation of oligomer from the polymer matrix, renders the coagent unavailable to subsequent polymer branching chemistry. However, retention of oligomerized coagent provides a monomer bearing multiple C=C groups, whose conversion to polymer grafts can produce a more extensive crosslink network than the trifunctional starting monomer.

The importance of the solubility of coagent-derived intermediates to LCB yields was tested in a series of small molecule experiments involving tetradecane as a model for EOC, and lactide as a model for PLA. The solubility limit of TAM at 170°C was determined to be 36 wt% (25 mol%) in tetradecane and 56 wt% (36 mol%) in lactide, as determined by  $^1\text{H-NMR}$  analysis of saturated solutions. Therefore, 5 wt% monomer solutions produced clear liquid mixtures from both model compounds. However, when a 5 wt% solution of TAM in tetradecane solution was heated with 1 wt% DCP, a homogeneous initial condition produced a white solid within one half-life of the initiator (See Supplemental Information), which is consistent with a previous report of sub-micron particle generation from this mixture [55]. Soxhlet extraction with xylenes recovered 4.5% solids from the crude product (Table 3), while FT-IR analysis confirmed the presence of both TAM and tetradecane within the crosslinked precipitate.

Table 3. Solids yield recovered from model compound grafting experiments

| Model Compound | Coagent | Solids Content (%) |
|----------------|---------|--------------------|
| Tetradecane    | TAM     | 4.5                |
| Tetradecane    | TMPTA   | 4.1                |
| Lactide        | TAM     | 0.0                |
| Lactide        | TMPTA   | 0.0                |

[DCP] = 37  $\mu\text{mol}\cdot\text{g}^{-1}$ ; [Coagent] = 150  $\mu\text{mol}\cdot\text{g}^{-1}$

The same experiment using lactide in the place of tetradecane showed no change in clarity when the solution was heated beyond five peroxide half-lives.  $^1\text{H-NMR}$  analysis of the reaction product confirmed that at least 35% of the allylic functionality charged as TAM to the reaction mixture was converted, yet filtration through a

Soxhlet thimble and rinsing with chloroform at 50 °C recovered no solid phase. This confirms that peroxide-initiated graft modification of lactide with TAM does not result in a precipitation polymerization, but remains as a homogeneous solution. Note that all attempts to recover crosslinked particles from PLA derivatives failed to isolate a solid phase. As a result, we conclude that remarkable efficiency of PLA branching is due to the retention of highly functional oligomers by the polymer matrix. This phase stability advantage outweighs the marginal grafting reactivity of PLA by allowing all coagent charged to a formulation to be utilized for LCB production.

#### 4. Conclusion

Radical-mediated coagent chemistry introduces long chain branching to PLA much more efficiently than analogous polyolefin modifications. This unique sensitivity to allylic coagents is not a result of ionic chemistry or an efficient radical grafting mechanism. Indeed, PLA and related polyesters with low aliphatic hydrocarbon content are relatively unreactive toward radical graft modification, providing relatively low AE and monomer graft yields due to a dearth of reactive H-atom donors. The difference between PLA and EOC appears to be thermodynamic solubility of coagent oligomers. Hansen solubility parameters confirm that PLA has a greater affinity for allylic esters and acrylates. More importantly, the graft modification of tetradecane, a model for EOC, loses allylic coagent to a precipitation polymerization process, whereas lactide, the monomer of PLA, maintains a miscible condition. This retention of coagent oligomers during the reaction supports more extensive long chain branching yields in spite of relatively inefficient radical grafting yields.

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## Supplemental Information

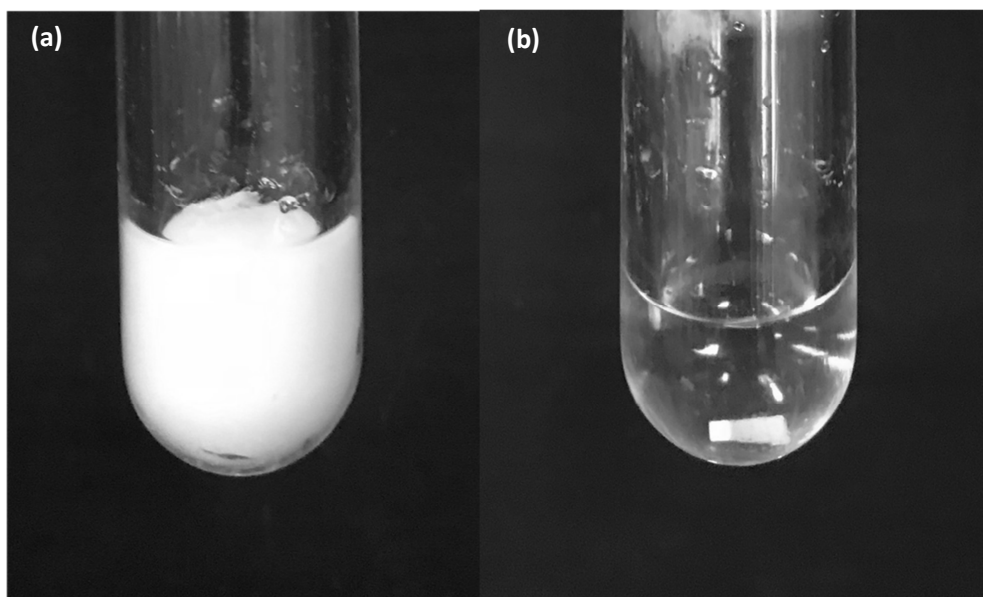


Figure S1: Photographs of products derived from DCP-initiated grafting of TAM to model compounds (a. tetradecane; b. lactide; [DCP]= 1 wt%; [TAM]= 5 wt%; 170°C; 5 min)

**Peroxide-initiated Graft Modification of Thermoplastic BioPolyesters:****Introduction of Long-Chain Branching**

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**Highlights**

- Coagent branching processes are more efficient for PLA than polyolefins
- PLA is a poor H-atom donor and a low-yield grafting substrate
- PLA's ability to solvate coagent oligomers underlies its branching reactivity